

Continuous functions



Outline

Algebra of continuous functions theorem

Continuity of elementary functions theorem

Continuity of composite functions theorem

Algebra of continuous functions theorem



Introduction: algebra of continuous functions

- ▶ In this section we see how **continuity** behaves with respect to the algebraic operations: **sum**, **difference**, **product** and **quotient**
- ▶ The result is a direct consequence of the **theorem on the algebra of limits**
- ▶ It will allow us to recognize the continuity of complicated functions starting from that of simpler functions

Theorem 1: Algebra of continuous functions

Let f and g be two functions defined at least in a neighborhood of $x_0 \in \mathbb{R}$ and continuous at x_0 . Then:

- 1. $f(x) \pm g(x)$ is continuous at x_0 ;*
- 2. $f(x) \cdot g(x)$ is continuous at x_0 ;*
- 3. $\frac{f(x)}{g(x)}$ is continuous at x_0 provided that $g(x_0) \neq 0$.*

Proof

Let us prove, for example, claim 3., the others being analogous. By hypothesis we know that f and g are continuous at x_0 , that is:

$$f(x) \rightarrow f(x_0) \quad \text{and} \quad g(x) \rightarrow g(x_0) \quad \text{as} \quad x \rightarrow x_0$$

Moreover $g(x_0) \neq 0$ and hence, by the sign-preservation theorem for continuous functions, $g(x) \neq 0$ eventually as $x \rightarrow x_0$.

Then, by the theorem on the algebra of limits, we conclude that

$$\frac{f(x)}{g(x)} \rightarrow \frac{f(x_0)}{g(x_0)} \quad \text{as} \quad x \rightarrow x_0$$

that is, $f(x)/g(x)$ is continuous at x_0 . □

Algebra of continuous functions:

- ▶ Sum, difference and product of functions continuous at x_0 are continuous at x_0
- ▶ The quotient is continuous at x_0 provided that the denominator does not vanish at x_0
- ▶ The proof uses the theorem on the algebra of limits and sign preservation

Continuity of elementary functions theorem

Introduction: continuity of elementary functions

- ▶ In this section we see that the **elementary functions** (powers, exponentials, logarithms, sine and cosine) are **continuous** on their domain
- ▶ We prove in detail the continuity of $\sin x$ and $\cos x$ on the whole of $\mathbb{R}$
- ▶ Together with the algebra of continuous functions theorem, we will obtain the continuity of polynomials, rational functions, $\tan x$ and $\cot x$

Continuity of elementary functions

Theorem 2: Continuity of elementary functions

The following elementary functions are continuous at all points of their domain:

1. *Powers with integer, rational or real exponent;*
2. *Exponential functions;*
3. *Logarithmic functions;*
4. *Elementary trigonometric functions ($\sin x$, $\cos x$)*

Proof (1/4)

For example, let us prove the continuity on the whole of $\mathbb{R}$ of the functions $\sin x$ and $\cos x$.

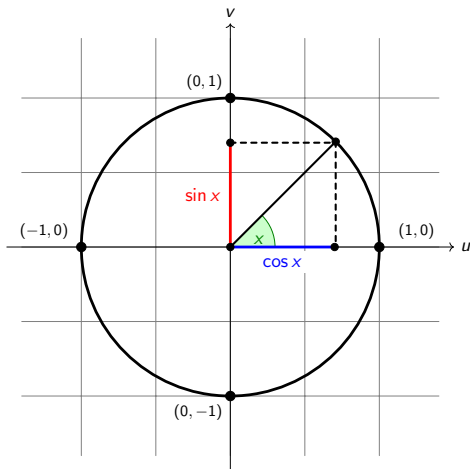
- ▶ We have seen that $\sin x$ is continuous at $x = 0$; let us show that $\cos x$ is also continuous at $x = 0$.
- ▶ The unit (trigonometric) circle shows that, if x is an angle in the first quadrant,

$$\sin x + \cos x \geq 1$$

since 1 is the hypotenuse of a right triangle with legs $\sin x$, $\cos x$.



Proof (2/4)



Proof (3/4)

- It follows that

$$0 \leq 1 - \cos x \leq \sin x \quad \text{for } x \in \left[0, \frac{\pi}{2}\right], \quad \text{and hence}$$

$$0 \leq 1 - \cos x \leq |\sin x| \quad \text{for } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

- Then, by the comparison theorem,

$$1 - \cos x \rightarrow 0 \quad \text{as } x \rightarrow 0 \quad \text{hence} \quad \cos x \rightarrow 1 \quad \text{as } x \rightarrow 0$$

and therefore $\cos x$ is continuous at 0.

Proof (4/4)

To now prove the continuity of $\sin x$ at a generic point $x_0 \in \mathbb{R}$ we consider the chain of relations:

$$\begin{aligned} |\sin(x_0 + h) - \sin x_0| &= |\sin x_0 \cos h + \sin h \cos x_0 - \sin x_0| \\ &= |\sin x_0 (\cos h - 1) + \cos x_0 \sin h| \\ &\leq |\sin x_0 (\cos h - 1)| + |\cos x_0 \sin h| \\ &= |\sin x_0| |\cos h - 1| + |\cos x_0| |\sin h| \end{aligned}$$

Now $|\sin h|$ and $|\cos h - 1|$ tend to zero as $h \rightarrow 0$, by what we have just proved, while $|\sin x_0|$ and $|\cos x_0|$ are constants, so

$$\sin(x_0 + h) - \sin x_0 \rightarrow 0 \quad \text{that is} \quad \sin(x_0 + h) \rightarrow \sin x_0 \quad \text{as} \quad h \rightarrow 0$$

and $\sin x$ is continuous at x_0 . An analogous argument shows the continuity of $\cos x$. □

Consequences

- ▶ Having proved that $\sin x$, $\cos x$ are continuous on the whole of $\mathbb{R}$, we deduce that the functions $\tan x$, $\cot x$ are continuous on their domain (by the theorem on the algebra of continuous functions)
- ▶ Power functions with integer exponent are continuous (on the whole of $\mathbb{R}$), since $f(x) = x$ is obviously continuous, and $f(x) = x^n$ is the product of n continuous functions.
- ▶ Polynomials are continuous functions, since they are obtained by adding functions of the type $c x^n$, which are continuous by the previous point.
- ▶ Rational functions (i.e., quotients of polynomials) are continuous functions, except at the points where the denominator vanishes (the denominator is a polynomial, hence it vanishes at a finite number of points)

Summary

Elementary functions:

- ▶ Powers, exponentials, logarithms, $\sin x$ and $\cos x$ are continuous on their domain
- ▶ The continuity of $\cos x$ at 0 follows from $0 \leq 1 - \cos x \leq |\sin x|$ and the comparison theorem
- ▶ The continuity of $\sin x$ at every x_0 follows from the addition formula

Consequences:

- ▶ $\tan x$, $\cot x$, polynomials and rational functions are continuous on their domain

Continuity of composite functions theorem



Introduction: continuity of composite functions

- ▶ In this section we see that the **composition** of continuous functions is still a continuous function
- ▶ The result follows from the **change of variable theorem** for limits
- ▶ We will derive a practical rule: for continuous functions the limit at a point of the domain is computed **by substitution**

Continuity of the composite function

Theorem 3: Continuity of the composite function

Let:

- ▶ *g be a function defined at least in a neighborhood of x_0 and continuous at x_0 ,*
- ▶ *f be a function defined at least in a neighborhood of $t_0 = g(x_0)$ and continuous at t_0 ,*

then $f \circ g$ is defined at least in a neighborhood of x_0 and is continuous at x_0 .

Proof

Since g is continuous at x_0 ,

$$\lim_{x \rightarrow x_0} g(x) = g(x_0) = t_0$$

then, by the change of variable theorem for limits, we have:

$$\lim_{x \rightarrow x_0} f(g(x)) = \lim_{t \rightarrow t_0} f(t)$$

and since f is continuous at t_0 , we have

$$\lim_{t \rightarrow t_0} f(t) = f(t_0)$$

and the thesis is proved. □

Combining continuous functions

- ▶ It follows that all the functions that can be obtained from elementary functions by sums, products, quotients and compositions are continuous on their domain. **Hence, by combining functions in this way, we still obtain continuous functions.**

To summarize:

- ▶ sums, products and quotients of continuous functions give continuous functions (where the denominator does not vanish);
- ▶ the elementary functions of mathematical analysis are continuous on their domain;
- ▶ the composition of continuous functions gives a continuous function.

Limits of continuous functions

- It is therefore possible to know a priori that a function is continuous on its domain, without applying the definition of continuity case by case.

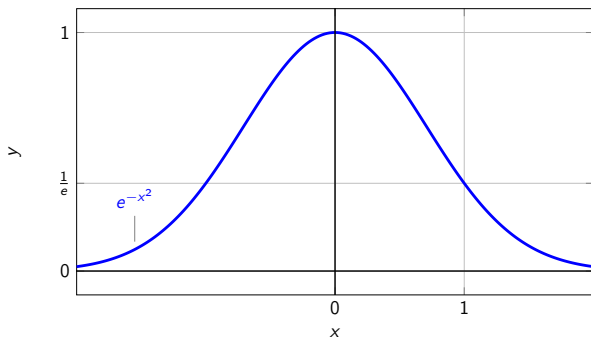
For continuous functions, if x_0 is a point of the domain, the limit as $x \rightarrow x_0$ is computed simply by substituting x_0 into the analytic expression of the function, that is:

$$f(x) \rightarrow f(x_0) \quad \text{as } x \rightarrow x_0$$

Example (1/3)

The following function is continuous for $x \in \mathbb{R}$:

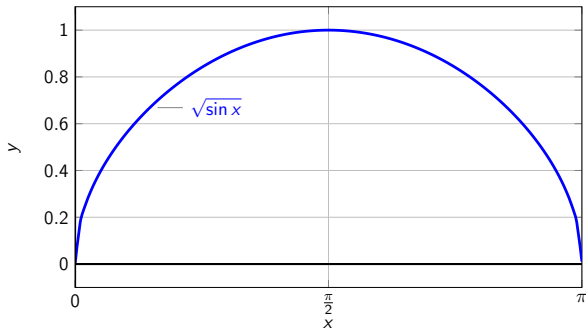
$$f(x) = e^{-x^2} \quad \text{hence} \quad \lim_{x \rightarrow 1} e^{-x^2} = e^{-1} = \frac{1}{e}$$



Example (2/3)

The following function is continuous for $x \in \mathbb{R}$ with $2k\pi \leq x \leq 2k\pi + \pi, \forall k \in \mathbb{Z}$:

$$f(x) = \sqrt{\sin x} \quad \text{hence} \quad \lim_{x \rightarrow \pi^-} \sqrt{\sin x} = \sqrt{\sin \pi} = 0$$



Example (3/3)

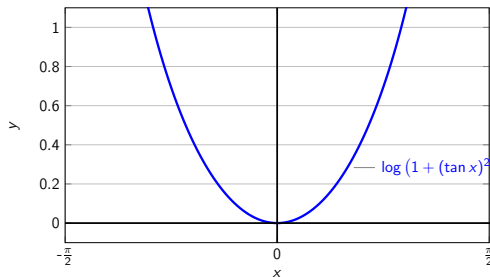
The following function is continuous for $x \in \mathbb{R}$ with $x \neq (2k+1)\frac{\pi}{2}, \forall k \in \mathbb{Z}$:

$$f(x) = \log_a (1 + (\tan x)^2), \quad \forall a > 0, a \neq 1$$

hence:

$$\lim_{x \rightarrow 0} \log_a (1 + (\tan x)^2) = \log_a (1 + (\tan 0)^2) = \log_a 1 = 0$$

For example, with $a = e$ we have:



Summary

Composite function:

- ▶ If g is continuous at x_0 and f is continuous at $g(x_0)$, then $f \circ g$ is continuous at x_0
- ▶ Sums, products, quotients and compositions of elementary functions are continuous on their domain

Practical rule:

- ▶ For a continuous function the limit at a point x_0 of the domain is computed by substitution: $f(x) \rightarrow f(x_0)$

Final summary

- ▶ **Algebra:** sum, product and quotient (with non-zero denominator) of continuous functions are continuous
- ▶ **Elementary functions:** powers, exponentials, logarithms, $\sin x$, $\cos x$ are continuous on their domain; consequently so are $\tan x$, $\cot x$, polynomials and rational functions
- ▶ **Composition:** the composition of continuous functions is continuous
- ▶ **Limits:** for continuous functions the limit at a point of the domain is obtained by substituting the point into the expression of the function